Mca project report

SAMIKSHA CHAVAN(23108A0057)|JAGRUTI THOTAM(23108A0066) |HIMANSHU SHINDE(23108A0068)

BURGLAR ALARM SYSTEM USING ARDUINO AND PIR SENSOR

PROBLEM STATEMENT:

Security is a major concern for homes, offices, and shops. Conventional systems are expensive and complex to install.

The problem is to *design a simple, low-cost, and efficient burglar alarm* that detects unauthorized movement and triggers an alarm or light to alert the owner.

OBJECTIVE:

To develop a motion-based burglar alarm system using Arduino that detects human presence via a *PIR (Passive Infrared) sensor* and activates a *buzzer and LED* for indication.

COMPONENTS USED:

components	description
Arduino Uno	Main microcontroller board
PIR Sensor	Detects human motion using infrared radiation
Buzzer	Produces alarm sound
Power Supply (5V or USB)	Powers the Arduino board
Jumper wires	Circuit connection

WORKING PRINCIPLE:

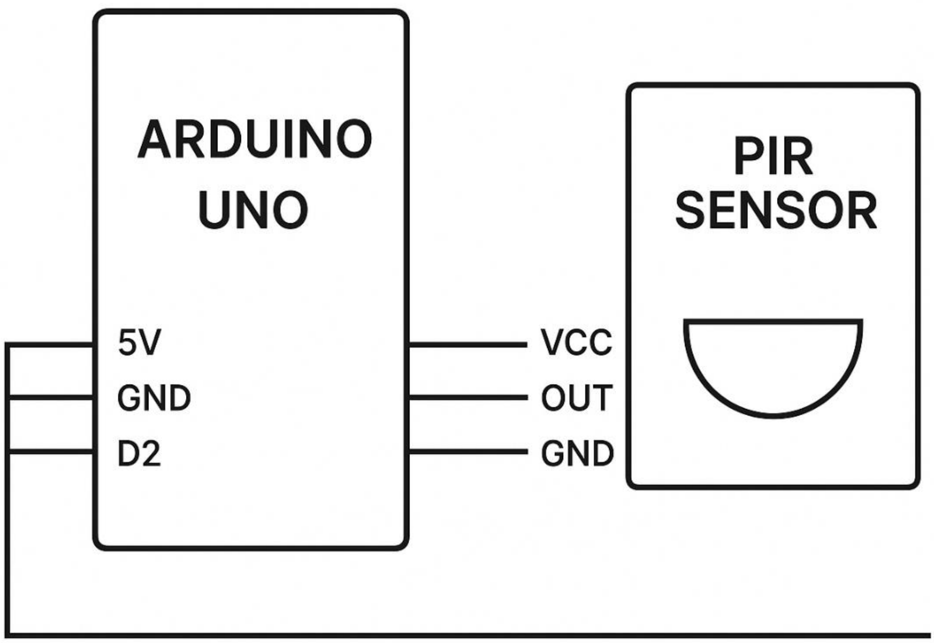
- The PIR sensor detects motion by sensing changes in infrared radiation emitted by humans.

- When the sensor detects movement, it sends a HIGH signal to the Arduino.
- Arduino processes this signal and activates the buzzer and LED.
- After a delay, the alarm resets and waits for the next detection.

PIR Sensor Working Principle:

- PIR sensors have two slots made of pyroelectric material.
- When both slots see the same IR levels (no motion), output is LOW.
- When a human (warm body) moves across, one slot detects higher IR radiation — the sensor output goes HIGH.
- This HIGH signal triggers the alarm.

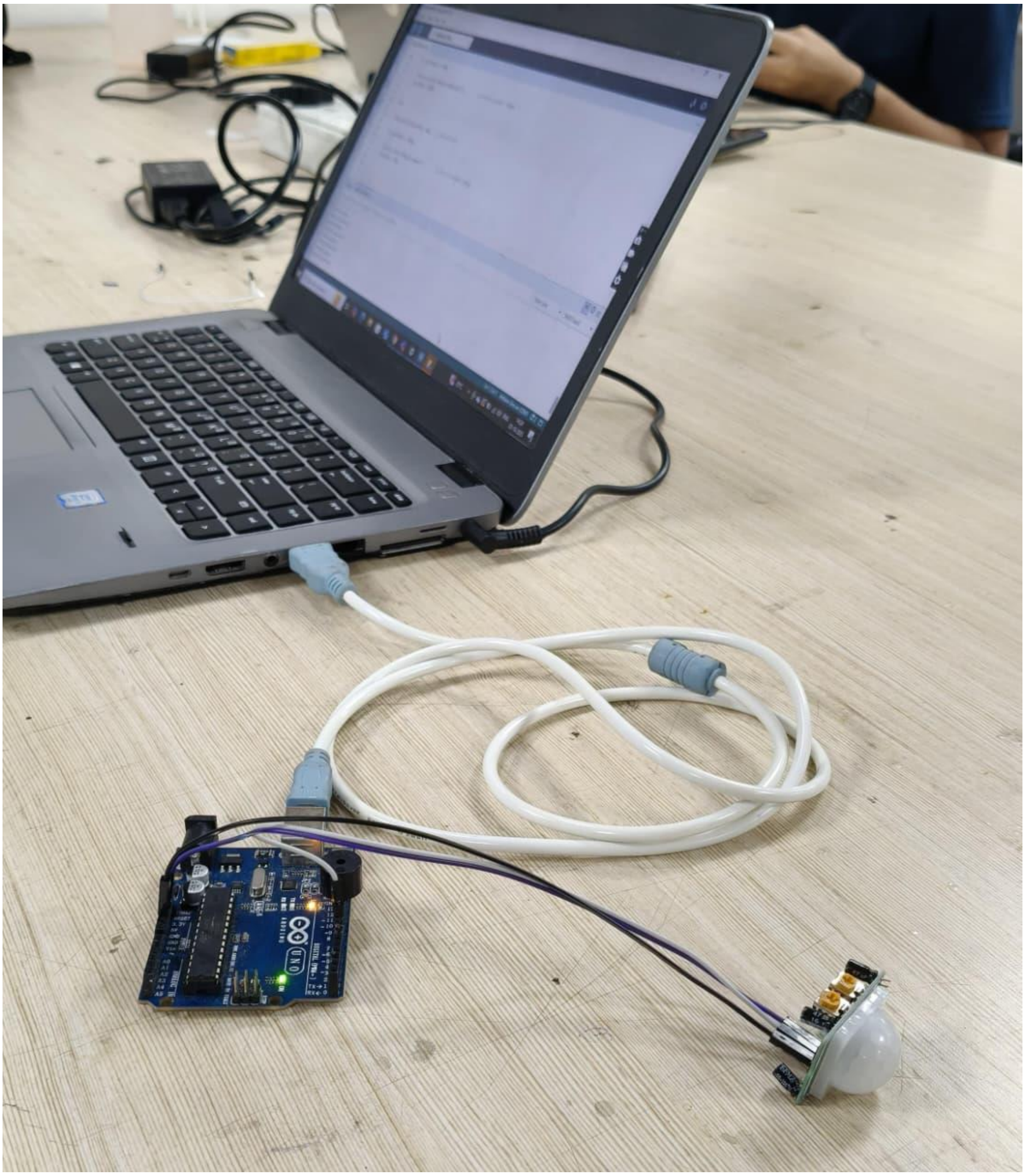
CIRCUIT DIAGRAM :



ARDUINO CODE:

```
int buzzerPin = 11;                 // choose the pin for the LED
int inputPin = 8;                   // choose the input pin (for PIR sensor)
int pirState = LOW;                 // we start, assuming no motion detected
int val = 0;                        // variable for reading the pin status

void setup() {
  pinMode(buzzerPin, OUTPUT);       // declare LED as output
  pinMode(inputPin, INPUT);         // declare sensor as input

  Serial.begin(9600);
}

void loop(){
  val = digitalRead(inputPin);      // read input value

  if (val == HIGH)                  // check if the input is HIGH
  {
    digitalWrite(buzzerPin, HIGH);  // turn LED ON

    if (pirState == LOW)
        {
      Serial.println("Motion detected!");        // print on output change
      pirState = HIGH;
    }
  }
  else
  {
    digitalWrite(buzzerPin, LOW);   // turn LED OFF

    if (pirState == HIGH)
        {
      Serial.println("Motion ended!");        // print on output change
      pirState = LOW;
    }
  }
}
```

PHOTO OF PROJECT:



APPLICATIONS:

- Home and office security systems
- ATM and bank vault monitoring
- Motion detection in restricted areas
- Smart door automation